Physics Department
Fisika Departement

Semester Test 2 *Semestertoets 2*

FSK116

9 May 2013 / *9 Mei 2013*

Surname and initials ______________________ Student number ____________________
Van en voorletters *Studentenommer*

Signature ______________________
Handtekening

Time *Tyd*	2½ hours *2½ ure*	Max. Total *Maks. Totaal*	70 Marks Max *70 Punte Maks*
Internal Examiner: *Interne Eksaminator*	Dr J M Nel	External Examiner *Eksterne Eksaminator*	Mr R Q Odendaal

• Answer all questions. • Cleary show all calculations. Do not just write down the answer. • Clearly state all assumptions, rules and laws used. • Where applicable draw the free-body diagrams. • Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. • Programmable calculators may NOT be used. • **NO** textbooks allowed. • A formula sheet is attached. • All test and examination regulations of the University of Pretoria are applicable.	• *Beantwoord alle vrae.* • *Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.* • *Stel duidelik alle aannames, reëls en wette wat u gebruik.* • *Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).* • *Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.* • *Programmeerbare sakrekenaars MAG NIE gebruik word nie.* • ***GEEN** handboeke word toegelaat nie.* • *'n Formuleblad is aangeheg.* • *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

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Semester Test 2 **FSK 116** **Semestertoets 2**

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Handtekening *Lokaal*

$v_{avg} = ½(v_0 + v)$ $v = v_0 + at$ $x = x_0 + v_0 t + ½at^2$ $x = x_0 + v_{avg} t$ $v^2 = v_o^2 + 2a(x - x_0)$ $x - x_0 = ½(v_0 + v)t$ $x - x_0 = vt - ½at^2$ $a = \frac{v^2}{r}.$ $g = 9{,}8\ \text{m/s}^2$ $h = \frac{v_0^2 \sin^2\theta}{2g}$ $R = \frac{v_0^2}{g}\sin 2\theta_0$	$T = \frac{2v_0 \sin\theta}{g}$ $f = \mu N$ $D = ½C\rho A v^2$ $W = \vec{F}\cdot\vec{d} = Fd\cos\phi$ $F = -kx$ $W = -½kx^2$ $P_{avg} = \frac{W}{\Delta t}$ $P = \frac{dW}{dt}$ $P = \vec{F}\cdot\vec{v}$ $U = ½kx^2$	$x_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i$ $\vec{r}_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i \vec{r}_i$ $\vec{p} = m\vec{v}$ $\vec{F}_{net} = \frac{d\vec{p}}{dt}$ $\Delta\vec{p} = \vec{J}$ $\vec{J} = \int_{t_i}^{t_f} F(t)\,dt$ $J = F_{avg}\Delta t$ $v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i}$ $v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$

Question 1 (Ch 5 – 8 , 5 – 41, 6 – 57) **6 + 10 + 7**
23

a) A 1.5 kg object is subjected to three forces that give it an acceleration $\vec{a} = \left[-(8.00)\hat{\text{i}} + (6.00)\hat{\text{j}}\right]\text{m/s}^2$. If two of the three forces are $\vec{F}_1 = \left[(30.0)\hat{\text{i}} + (16.0)\hat{\text{j}}\right]\text{N}$ and $\vec{F}_2 = \left[-(12.0)\hat{\text{i}} + (8.00)\hat{\text{j}}\right]\text{N}$, determine the third force.

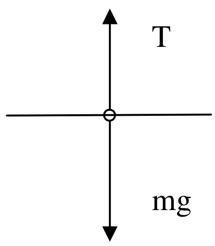
'n 1.5 kg voorwerp word onderwerp aan drie kragte wat 'n versnelling van $\vec{a} = \left[-(8.00)\hat{\text{i}} + (6.00)\hat{\text{j}}\right]\text{m/s}^2$ *gee. Indien twee van die drie kragte* $\vec{F}_1 = \left[(30.0)\hat{\text{i}} + (16.0)\hat{\text{j}}\right]\text{N}$ *en* $\vec{F}_2 = \left[-(12.0)\hat{\text{i}} + (8.00)\hat{\text{j}}\right]\text{N}$ *is, bepaal die derde krag.* [6]

$F_{net} = ma = 1.5\,\text{kg}\left(-\left(8.00\,\hat{\text{i}}\right) + \left(6.00\,\hat{\text{j}}\right)\text{m/s}^2\right)$ $= \left[-\left(12.0\,\hat{\text{i}}\right) + \left(9.00\,\hat{\text{j}}\right)\right]\text{N}$	✓✓
$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \left[-\left(12.0\,\hat{\text{i}}\right) + \left(9.00\,\hat{\text{j}}\right)\right]\text{N}$ $F_x = -12.0 = 30.0 - 12.0 + F_{3x}$ $F_{3x} = -12.0 - 30.0 + 12.0 = -30.00\,\text{N}$ $F_y = 9.00 = 16.0 + 8.00 + F_{3y}$ $F_{3y} = 9.00 - 16.0 - 8.00 = -15.0\,\text{N}$	✓ ✓

$\vec{F}_3 = \left[-\left(30.0\hat{\mathrm{i}}\right)-\left(15.0\hat{\mathrm{j}}\right)\right]\mathrm{N}$	✓✓

b) Using a rope that will snap if the tension in it exceeds 387 N, you need to lower a bundle of roof tiles weighing 449 N from a point 6.1 m above the ground.
Deur'n tou te gebruik wat sal breek indien die spankrag 387 N oorskry, moet jy 'n bondel dakteëls, wat 449 N weeg, laat sak vanaf 'n hoogte van 6.1 m bo die grond.

i) What magnitude of the bundle's acceleration will put the rope on the verge of snapping?
Wat moet die versnelling van die stapel teëls wees sodat die tou net-net nie breek nie? [2+3]



✓

✓

The mass of the bundle is m = (449 N)/(9.80 m/s^2) = 45.8 kg and we choose +y upward. The maximum tension allowed without snapping the rope is 387 N. Newton's second law, applied to the bundle, leads to $\sum F_y = ma = T - mg$ $\Rightarrow \quad a = \dfrac{387\ \mathrm{N} - 449\ \mathrm{N}}{45.8\ \mathrm{kg}} = -1.4\ \mathrm{m/s^2}$ i.e. 1.4 m/s^2 downwards is the minimum acceleration and would put the rope on the verge on snapping.	✓ ✓ ✓✓

ii) What will happen if the acceleration is less than this value in (a)?
Wat sal gebeur as die *versnelling minder is as die waarde in (a)?* [2]

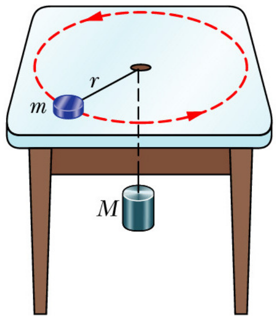
Below this accelleration the tension will be higher (beyond breaking point) and the rope will snap.	✓✓

iii) At this acceleration, at what speed would the bundle hit the ground?
Teen hierdie versnelling, wat sal die spoed van die stapel wees wanneer dit die grond tref? [3]

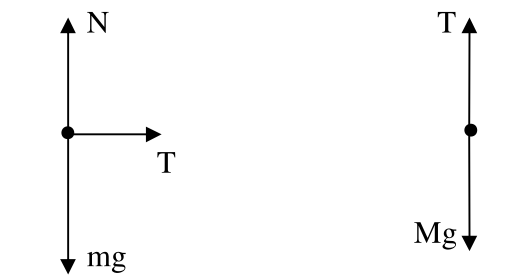
The bundle starts from rest and accelerates downwards with the value determined in (a) above. Δy = -6.1 m, v_0 = 0m/s, a = -1.4 m/s^{2,}	✓

$v^2 = v_0^2 + 2a_y\Delta y$ $v = \sqrt{2a_y\Delta y} = \sqrt{2\left(-1.4\,\text{m/s}^2\right)\left(-6.1\,\text{m}\right)}$ $= 4.1\ \text{m/s}$	✓✓

c) A puck of mass m = 1.50 kg slides in a circle of radius r = 20.0 cm on a frictionless table while attached to a hanging cylinder of mass M = 2.50 kg by a cord through a hole in the table. Show that the speed of the puck must be 1.81 m/s for the cylinder to be at rest.
'n Skuif met massa m = 1.50 kg beweeg in 'n sirkel met radius r = 20.0 cm op 'n wrywinglose tafel, terwyl 'n silinder met massa M = 2.50 kg aan 'n koord hang wat deur 'n gaatjie in die tafel gevoer is. Toon aan dat die spoed van die skuif 1.81 m/s moet wees om te die silinder te laat stilhang.

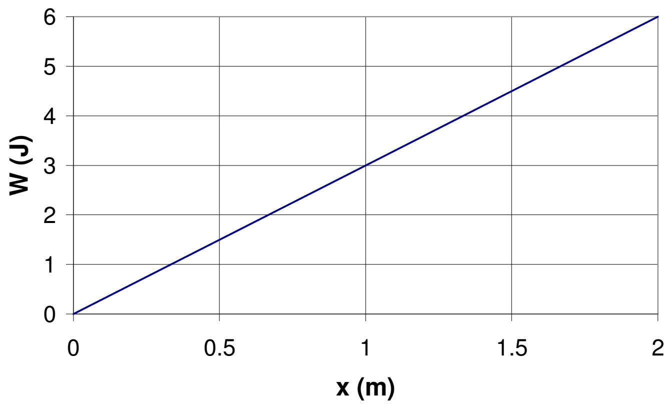


[2+5]

	✓✓
For puck mass *m:* $\sum F_y = ma_y = 0$ $0 = N - mg$ $\sum F_x = ma_x = m\frac{v^2}{r} = T$ For suspended mass *M:* $\sum F_y = Ma_y = 0$ $0 = T - Mg$ $0 = m\frac{v^2}{r} - Mg$ $v = \sqrt{\frac{Mgr}{m}}$ $= \sqrt{\frac{\left(2.50\,\text{kg}\right)\left(9.8\,\text{m.s}^{-2}\right)\left(0.20\,\text{m}\right)}{1.5\,\text{kg}}}$ $= 1.81\,\text{m/s}$	✓ ✓ ✓ ✓ ✓

Question 2 (Ch 7 – 12, Theory, 8-24, 8-39, 8-15) **6 + 7 + 10 + 10**

a) A box is pushed 2 m along an *x* axis along a greasy frictionless floor. The figure gives the work *W* done on the box by a constant horizontal force versus the box's position *x*.
'n Houer word 2 m langs 'n x-as oor 'n olierige wrywinglose vloer gestoot. Die figuur dui die arbeid W verrig op die houer deur 'n konstante horisontale krag teenoor die houer se posisie x.



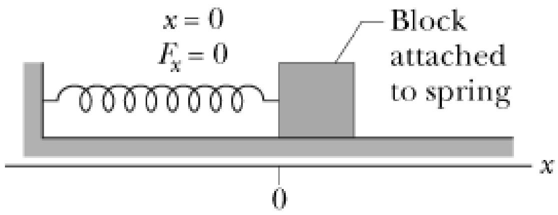
i) Determine is the magnitude of that force.
Bepaal die grootte van dié krag. [3]

Since $W = Fx\cos\phi$ and $\phi = 0°$ we find that $W \propto x$ the slope of the graph represents the constant force applied.	✓
$F = \frac{\Delta y}{\Delta x} = \frac{6}{2} = 3\,\text{N}$	✓✓

ii) If the box had an initial kinetic energy of 3 J, moving in the positive direction of the *x* axis, what is its kinetic energy after the 2 m displacement?
Indien 'n houer 'n aanvanklike kinetiese energie van 3 J gehad het, en in die positiewe rigting van die x as beweeg het, wat is die houer se kinetiese energie na die 2 m verplasing? [3]

From the Kinetic Energy Work Theorem, we get: $W = \Delta K$	✓
The total work done is 6 J (from the graph), therefore $K_f - K_i = W$	
$K_f = K_i + W = 3\text{ J} + 6\text{ J} = 9\text{ J}$	✓✓

b) Derive the general expression to determine the work done by a massless, ideal spring as it moves a block of mass *m*, through a distance *d*. Start from the basic definition of work: $W = \vec{F} \cdot \vec{d} = Fd\cos\phi$. Assume that the block is on a frictionless surface.

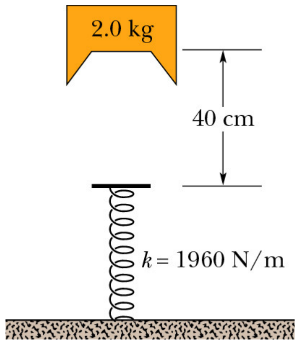


Lei die algemene vergelyking vir die bepaling van die arbeid verrig deur 'n massalose, ideale veer, soos dit 'n blok met massa m, deur 'n afstand d beweeg, af. Begin met die basiese definisie vir arbeid:- $W = \vec{F} \cdot \vec{d} = Fd\cos\phi$. Neem aan dat die blok op 'n wrywinglose oppervlak beweeg.

[7]

For a spring $\vec{F} = -k\vec{d}$ which is a variable force.	✓
Cannot apply above formula for work directly in $W = \vec{F} \cdot \vec{d} = Fd\cos\phi$, since this is only applicable for a constant force. We divide the distance into small increments Δx over which F can be considered a constant. The work done for one of these Δx intervals is then $\Delta W_i = F\Delta x$ The total amount of work done over the whole distance (x_i *to* x_f) is then the sum of the work done over all the Δx will then give the total work, $W = \sum F\Delta x$. In the limit $\Delta x \to 0$ we can write:	✓ ✓ ✓
$W = \int_{x_i}^{x_f} F(x)\,dx$ then substituting the function for force due to a spring - $\vec{F} = -kx$ $W = \int_{x_i}^{x_f} F(x)\,dx$ $= \int_{x_i}^{x_f} -kx\,dx = -k\int_{x_i}^{x_f} (x)\,dx$ $= -k\left[½x^2\right]_{x_i}^{x_f}$ $W = ½kx_i^2 - ½kx_f^2$	✓ ✓ ✓

c) A block of mass m = 2.0 kg is dropped from a height h = 40 cm onto a spring of spring constant k = 1960 N/m (See figure).
'n Blok met 'n massa m = 2.0 kg word van 'n hoogte h = 40 cm gelos en val op 'n veer met veerkonstante k = 1960 N/m (Sien figuur).



i) Determine the velocity of the block when it hits the spring.
Bepaal die snelheid van die blok wanneer dit die veer tref.

[4]

Let the reference point be the top of the spring. i.e. U = 0 J at top of spring. ✓ ✓

$$(K+U)_i = (K+U)_f$$ ✓

$$(0+mgh) = (½mv^2+0)$$ ✓

$$v = \sqrt{2mgh/m} = \sqrt{2gh}$$

$$= \sqrt{2(9.8\text{ m/s}^2)(0.4\text{ m})}$$ ✓

$$= 2.8\text{ m/s downwards}$$

can also use the equation:

$$v^2 = v_0^2 + 2a(\Delta y)$$

ii) Determine the maximum distance the spring is compressed.
Bepaal die maksimum afstand waarmee die veer saamgepers word. [3]

KE at top of spring = PE of spring at full compression.

$$\Delta K = \Delta U_{spring} + \Delta U_g$$ ✓

$$½mv^2 = ½kx^2 + mgh \qquad \text{where } h = -x$$

$$0 = kx^2 - 2mgx - mv^2$$

Using the formula for the quadratic solution ✓

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-(-2mg) \pm \sqrt{(-2mg)^2 - 4(k)(-mv^2)}}{2k}$$

$$= \frac{(2(2.0\text{kg})(9.8\text{m.s}^2)) \pm \sqrt{(-2(2.0\text{kg})(9.8\text{m.s}^{-2}))^2 - 4(1960\text{N/m})(-(2.0\text{kg})(2.8\text{m.s}^{-1})^2)}}{2(1960\text{N/m})}$$

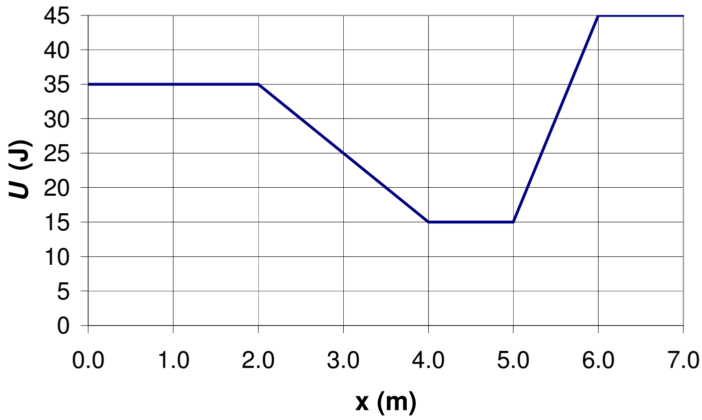
$$= 0.10\text{ m}$$ ✓

We chose the positive root, since we have already taken the compression into account (h = -x).

iii) Determine the work done on the block by the spring.
Bepaal die arbeid verrig deur die veer op die blok. [3]

The work done by the spring:	✓
$W = \int_0^x F dx = -\int_0^x kx\, dx = -½kx^2$	✓
$= -½(1960 \text{ N/m})(0.09 \text{ m})^2$	✓
$= -7.94 \text{ J}$	

d) The figure shows a plot of potential energy U versus position x of a 0.90 kg particle that can travel only along an x axis. (Non-conservative forces are not involved.) The particle is released at $x = 4.5$ m with an initial speed of 7.0 m/s, headed in the negative x direction.
Die figuur dui 'n grafiek van potensiële energie U teenoor posisie x van 'n 0.90 kg deeltjie wat sleg langs die x-as kan beweeg. (Nie-konserwatiewe kragte is nie betrokke nie.) Die deeltjie word by x = 4.5 m losgelaat met 'n aanvanklike spoed van 7.0 m/s, gerig in die negatiewe x-rigting.



i) If the particle can reach $x = 1.0$ m, what is its speed there, and if it cannot, what is its turning point?
Indien die deeltjie x = 1.0 m kan bereik, wat is sy spoed daar, en indien dit nie kan nie, waar is die draaipunt? [4]

$K = ½mv^2$ $= ½(0.90\,\text{kg})(7.0\,\text{m/s})^2$ $= 22.05\ \text{J}$	✓
The total energy E = U + K = 15.0 + 22.05 = 37.05 J	✓
This means the particle will reach $x = 1.0$ m since the potential energy there is 35 J leaving 2.0 J of kinetic energy and the particle will still be moving. $v = \sqrt{\frac{2K}{m}} = 2.1\,\text{m/s}$	✓ ✓

ii) What is the magnitude and direction of the force on the particle as it begins to move to the left of $x = 4.0$ m?
Wat is die grootte en rigting van die krag wat op die deeltjie inwerk wanneer dit begin om links van x = 4.0 te beweeg? [3]

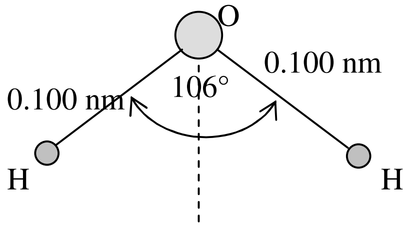
The gradient of the U vs x graph , $\frac{dU}{dx}$, is representative of the corresponding Force curve. This results in the force on the particle as it moves left of x = 4.0 m will be: $F = \frac{dU}{dx} = \frac{\Delta U}{\Delta x}$ $= \frac{20}{2} = 10 \text{ N}$	✓ ✓ ✓

iii) Suppose, instead, the particle is headed in the positive direction when released at x = 4.5 m at a speed of 7.0 m/s, will the particle reach x = 7.0 m? Clearly show your reasoning.
Veronderstel dat die deeltjie nou in die positiewe rigting beweeg, nadat dit losgelaat is by x = 4.5 m teen 'n spoed van 7.0 m/s, sal die deeltjie x = 7.0 m bereik? Verduidelik jou redenasie. [4]

The particle has the same energy as in i) a total of 37.0 J. The energy at x = 7.0 m is 45.0 J. The particle will therefore not reach x = 7.0 m. The turning point will be where the total energy = potential energy of 37 J, (between 5.5 m and 6.0 m)	✓ ✓ ✓ ✓

Question 3 (ch 9 - **6**

a) A water molecule consists of an oxygen atom and two hydrogen atoms bound to it as shown in the figure. The angle between the two hydrogen bonds is 106°. If the bond lengths are 0.100 nm long, determine the position of the centre of mass of the molecule. An oxygen atom is 16 times as heavy as a hydrogen atom.
'n Water molekule bestaan uit 'n suurstof atoom en twee waterstof atome wat gebind word soos aangedui in die figuur. Die hoek tussen die twee waterstof bindings is 106°. Indien die bindingslente 0.100 nm lank is, bepaal die posisie van die massa middelpunt van die molekule. 'n Suurstof atoom is 16 keer so swaar as 'n waterstof atoom. [5]



You need to clearly show your co-ordinate system (x- and You need to clearly show your co-ordinate system (x- and y- axes) Let the oxygen atom be at the origin of a conventional xy co-ordinate system. Due to symmetry, the horizontal component of the COM will be at x = 0 nm	✓ ✓
For the y axis, the hydrogen atoms are at a distance of:	✓

$y = r\cos 53° = 0.1(0.602) = 0.0602\,\text{nm}$ below y = 0 nm. This results in the COM in the y-direction: $y_{com} = \frac{1}{M}\left[m_O x_O + m_{H1}x_{H1} + m_{H1}x_{H1}\right]$ $= \frac{1}{18\,\text{H}}\left[(16\,\text{H})(0\,\text{nm}) + (1\,\text{H})(-0.0602\,\text{nm}) + (1\,\text{H})(-0.0602\,\text{nm})\right]$ $y_{com} = -0.00689$ nm	✓ ✓
COM of water molecule is x_{com} = 0 nm and $y_{com} = -0.00689$ nm on the coordinate system used. Depending on your co-ordinate system – your x_{com} must have the same position as your oxygen atom, and your y_{com} needs to be ~0.007 nm below the oxygen atom.	

b) A vessel at rest at the origin of an xy co-ordinate system explodes into three pieces. Just after the explosion, one piece of mass m, moves with a velocity $\left(-30\,\text{m.s}^{-1}\right)\hat{\imath}$ and the second piece, also of mass m, moves with velocity $\left(-30\,\text{m.s}^{-1}\right)\hat{\jmath}$. The third piece has mass $3m$. Just after the explosion, what are the magnitude and direction of the velocity of the third piece?
'n Houer wat aanvanklik met rus lê by die oorsprong van 'n xy -koördinaatstelsel ontplof in drie stukke. Net na die ontploffing beweeg die een stuk, met massa m, met 'n snelheid $\left(-30\,\text{m.s}^{-1}\right)\hat{\imath}$ *en die tweede stuk, ook met massa m, met 'n snelheid* $\left(-30\,\text{m.s}^{-1}\right)\hat{\jmath}$. *Die derde stuk het 'n massa 3m. Bereken die grootte en rigting van die snelheid van die derde stuk net na die ontploffing.* [9]

Conservation of momentum: $P_i = P_f$ In the x-direction: $p_{ix} = p_{f1x} + p_{f2x} + p_{f3x}$ $0 = (-30m) + 0 + p_{f3x}$ $p_{f3x} = 30m = 3mv_{f3x}$ $v_{f3x} = 10\,\text{m.s}^{-1}$	✓ ✓ ✓
In the y-direction: $p_{iy} = p_{f1y} + p_{f2y} + p_{f3y}$ $0 = 0 + (-30m) + p_{f3y}$ $p_{f3y} = 30m = 3mv_{f3y}$ $v_{f3y} = 10\,\text{m.s}^{-1}$	✓ ✓ ✓

$v_{3f} = \left(10\ \text{m.s}^{-1}\right)\hat{\text{i}} + \left(10\ \text{m.s}^{-1}\right)\hat{\text{j}}$ $\lvert v_{3f} \rvert = \sqrt{10^2 + 10^2} = 14\ \text{m.s}^{-1}$ $\theta = \tan^{-1}\left(\frac{10}{10}\right) = 45°$ Above the positive x-axis.	✓ ✓ ✓

Finish & Klaar